

From Sensorimotor Schemes to Collective Behavior: Developmental Inference of Other Agents' Behaviors and their Consequences on the Environment through Sensorimotor Interactions

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Abstract—Radical Interactionism (RI) is a formulation of the problem of self-discovery enabling artificial agents to infer, from sensorimotor schemes, the properties of space, build models of local environments, and discover the motives of other agents. RI agents are thus able to develop individual and collective behaviors satisfying inborn motivational drives, without a priori knowledge. A remaining challenge for developmental agents is the autonomous discovery of agent-environment dynamics necessary to anticipate outcomes and autonomously coordinate complex multi-agent tasks.

In this article, we extend a RI architecture to enable the inference of behavioral motives of other agents according to their surrounding contexts of affordances (static or mobile entities, including oneself), and learn the consequences of their actions on the environment. These new possibilities allow the emergence of collective behaviors of higher complexity, collaborative or competitive tasks, requiring the anticipation of further steps of a process. We illustrate our proposal with a working implementation in a competitive foraging multi-agent environment, where the agent learns to anticipate competitors' actions to adapt its resource-seeking strategy.

Index Terms—developmental learning, interactionism, affordance, behavior inference, autonomous mental development

I. INTRODUCTION

Among the Developmental learning [1] approaches aimed at creating agents able to interact with their environment without *a priori* knowledge, the Interactionist approach [2] assumes that the perception process is an internal construct resulting from active experience. This approach is in line with constructivism [3] and enactive learning [4], which assume that learning about the environment and the emergence of behaviors are built on the basis of *sensory-motor schemes* [5] or, in Interactionist terms, *interactions*.

The Radical Interactionist (RI) approach, illustrated in Fig. 1, relies exclusively on interactions. A RI agent discovers its coupling with the environment and constructs a persistent spatial model that can be exploited to generate self-motivated behaviors [6], and even infer behavior models of other agents [7]. However, to enable the emergence of collective behaviors, an agent must be able to predict the consequence of other agents' behaviors, such as displacing or removing other entities. Such ability is required for instance in hunting scenarios, either competitive (prey availability) or collaborative (circling or herding prey).

In this paper, we enable RI agents to discover and project the expected outcome of other agents' interactions in their model, leading to the emergence of finer collective behaviors. We illustrate our proposal with a working simulation example of competitive foraging, where agents favor food sources they may reach before competitors. Fig. 2 details how the proposed mechanisms fit in our RI architecture [7].

Section II reviews related work and summarizes core principles of the RI approach. Section III details mechanisms for the integration of complex agents and Section IV details mechanisms for learning and exploiting consequences of others' behaviors. Intermediate results are presented with the description of each sub-mechanism to provide an intuitive comprehension of their functioning. Section V concludes with future directions.

II. THE RADICAL INTERACTIONIST APPROACH

For the sake of brevity, the basic principles of RI are illustrated with examples from the discrete simulation environment used for this article's contribution, shown in Fig. 3. However, the RI approach is not limited to discrete environments [8]. In this 2D environment, fishes implemented as RI agents attempt to feed on static food sources (blue algae) and avoid collision with obstacles (green blocks). Distractors are also present (red algae). RI agents are given in-born preferences for some interactions (sensorimotor scheme), such as eating being satisfying or colliding being painful. The *valence* v_i of an interaction i is a numerical value representing such in-born preferences, supporting a form of intrinsic motivation [9] called *interactional motivation* [10]. Unlike rewards that are considered and optimized *a posteriori*, shaping the RL agent's inner model and behavior, the valences drive the decisions *a priori*: the RI agent only considers the reliability of interaction predictions of the inner environment model. This property makes the valence independent from the environment model, enabling the use of dynamic valence, for instance to modulate the valence of an 'eat' interaction according to hunger.

The sensorimotor apparatus of fishes allows the following interactions (valences in parentheses): move forward $\triangleright$ (2), collide with an obstacle $\blacktriangleright$ (-5), eat $\blacktriangleright$ (50), "cuddle" with other fishes $\blacktriangleright$ (5), turn right $\cup$ (-3) and left $\triangleleft$ (-3) by 90°.

Fishes have a forward 180° field of view and can perceive 7 visual stimuli: three colors (red, green and blue) and four orientations of other fishes. the visual system discretizes this space as a regular grid of 15x9 unlabeled positions.

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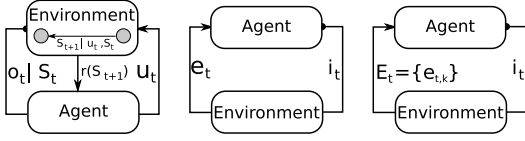


Fig. 1. Reinforcement learning (left): the agent is driven by a reward resulting from a state change. Radical Interactionism [2] (middle): the agent can only interact with its environment through sensorimotor schemes called interactions. The agent intends an interaction i_t and as a result experiences the enacted interaction e_t . Parallel RI [7] (right): an extension of RI in which the enacted feedback is a set of interactions.

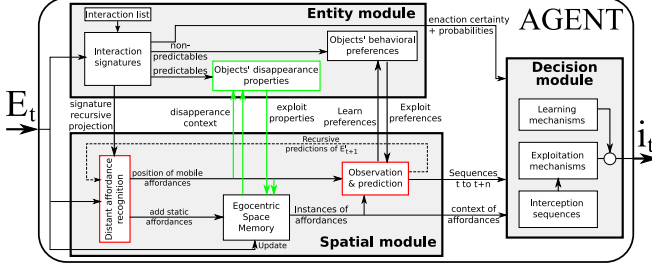


Fig. 2. This article extends our RI architecture [7], enabling it to integrate the consequences of other agents' interactions with the environment. The *Entity module* constructs models of affordances, the *Spatial module* learns to localize and keep track of affordances in extra-personal space, the *Decision module* exploits other modules to generate intrinsically motivated behaviors. The sub-modules in red are extended for the integration of complex agents. The new sub-module in green learns their effect on the environment.

We denote a *primary interaction* a couple (action, stimuli), that is, a sensorimotor scheme composed of an action and a perception *resulting* from this action.

The RI model proposes that, while enacting a primary interaction, the agent can experience additional stimuli. These stimuli must be considered with the movement produced by the enacted interaction. As an example, identical stimuli from an optical flow, produced from two different movements, may characterize entities at different distance. We denote a *secondary interaction* a couple (interaction, stimuli) that can only be enacted when its associated primary interaction is enacted. The RI agents of our simulation have a set I containing 6 primary interactions (move forward, bump...). Their visual system, combined with the 5 primary interactions producing a movement, adds a total of $15 \times 9 \times 7 \times 5 = 4725$ uninterpreted 'visual' secondary interactions.

When a RI agent [6] intends an interaction i_t (either primary or secondary), it experiences the set of interactions E_t which are actually enacted. The enaction is a success if $i_t \in E_t$ and a failure otherwise (e.g. *move* may fail because of an obstacle, E_t will contain *bump* instead). Also, all interactions in E_t succeeded. E_t contains one primary interaction and possibly a set of secondary interactions related to this primary.

A. Identifying experiences leading to a successful interaction

From an external point of view, the success of an interaction is related to the presence of a specific entity in the environment, that can be described as an *affordance*. A RI agent cannot directly perceive affordances, but can detect them through certain enacted interactions in E_t .

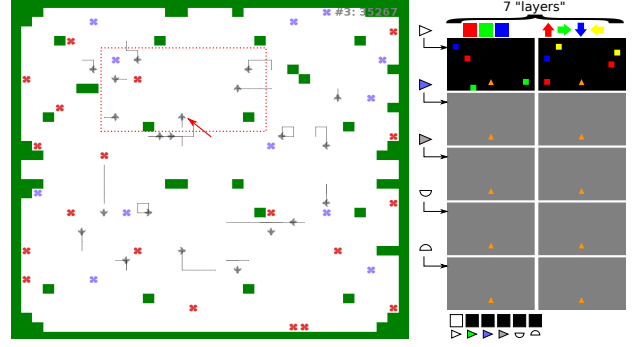


Fig. 3. Left: the test environment is populated with obstacles (green blocks), food sources (blue algae), distractors (red algae), and 20 agents (grey fishes). Agents respond to 7 visual stimuli corresponding to the 3 static elements and the 4 possible orientations on other agents. The red dotted rectangle shows the field of view of the agent designated by the red arrow. Right: the uninterpreted interactional system of an agent is organized as follows: visual interactions are gathered according to each movement interactions (shown as 5 rows). For each, interactions are organized to match their ground truth position in the visual field, with a layer for each of the 7 visual stimuli. Layers are overlaid in two columns: the 3 stimuli corresponding to static elements on the left, the 4 stimuli corresponding to agents' orientations on the right (the colors match the red, green, blue and yellow arrows). The figure shows the enacted context E_t of the agent highlighted on the left, which intended to move forward (indicated by the white square below), thus enacting the corresponding visual interactions (first row).

A *Signature* S_i of an interaction i is a data structure that implements a function $S_i : \mathcal{P}(I) \rightarrow [-1, 1]$ of context E_t predicting the certainty of success or failure of i at decision cycle $t+1$ ($S_i(E_t) = 1$ for absolute certainty of success, -1 for failure). The signature S_i is learned by identifying interactions whose enaction allows the prediction of the result of i , which is equivalent to detecting its affordance.

A signature S_i must define a pseudo-reverse function $\hat{S}_i$ designating several sets of interactions affording ($\hat{S}_i(1)$) or preventing ($\hat{S}_i(-1)$) the enaction of i , each set corresponds to a primary interaction j^p (e.g. move, turn ...) and a subset of associated secondary interactions $\{j_k^s\}$ (e.g. visual interactions) that can have their own signatures. By using signatures $S_{j_k^s}$ of interactions of such a subset $\{j_k^s\} \subset \hat{S}_i(1)$ related to the same primary interaction j^p , it becomes possible to define a context $S_i^{j^p}$ that, after enacting j^p , may afford i . This principle can be applied recursively through a sequence $\sigma = \langle j_1^p, \dots, j_{steps}^p \rangle$ of primary interactions, enabling the detection of distant affordances through the obtained "projected" signatures S_i^σ . Distant affordances are localized through shortest sequences leading to them [6].

The use of affordances [12], [11] as a means of representing the environment has been extensively studied (e.g. [13], [14], [15], but was limited to short term predictions due to their strong coupling with perception only. Using interactions to represent affordances, and the movements produced by interactions to localize them and maintain a spatial distant context dispenses with notions external to the agent, such as entities, or even the notion of space itself.

B. Managing and exploiting a local model of affordances

When a projected signature S_a^σ matches the current experienced context E_t , an affordance instance of a is stored in a mechanism called Egocentric Space Memory (ESM) [6]

at the position defined by the sequence σ . The ESM stores triplets (a, i, d) with a the afforded interaction, i the first interaction of σ (providing a "direction"), and d its distance (the length of the shortest σ). We denote $M_t = \{(a_k, i_k, d_k)\}$ the current context of affordance stored in the ESM M . The ESM learns to update the (a, i, d) triplets, keeping track of a context larger than its perceptual field, and implementing a form of object permanence.

Behaviors are generated based on context M_t and signatures' predictions. The agent favors enactable interactions (i.e. positive $S_i(E_t)$) bringing affordances of interactions with positive valences closer, and remain far from affordances of "negative" interactions. A utility value is defined for each interaction i predicted as a success:

$$u_i = \nu_i + \alpha \times \sum_{(a_n, i_n, d_n) \in M_t} \nu_{a_n} \cdot f(d_n) \cdot id(i, i_n) \quad (1)$$

where ν_i is the valence of i , ν_{a_n} the valence of a_n , f is a strictly decreasing and positive function, $id(i, i_n) = 1$ if $i = i_n$ and 0 otherwise. α weights the influence of distant affordances and interaction valences [6]. We use $f(d) = e^{-0.05 \cdot d}$ and $\alpha = 0.4$ in our simulation.

Another important function of the ESM is the simulation of reference changes through a given sequence σ , simulating the context from the point of view of another affordance, enabling distance computation between affordances [7]. We denote $M_t^\sigma = \{(a_k, i_k^\sigma, d_k^\sigma)\}$ the content of a simulated context at "position" σ .

III. DISCOVERING OTHER AGENTS

Affordances offered by mobile entities may move during the enaction of an interaction, therefore the prediction error of the corresponding signature cannot be reduced. Previous works [7] introduce *partial signatures* to deal with the probable position of a simple mobile entity. However, discovering and interacting with complex entities such as oriented agents require a finer model of their capabilities.

A. Learning signatures of mobile affordances

For a RI agent, an affordance can only be perceived through a set of previously enacted interactions designated by the signature of interaction. The signatures thus define relations between interactions, and can encode spatial properties such as the movement produced by a primary interaction, and, in the case of a mobile affordance, the multiple movements of the affordance [7].

In this article, a signature is implemented as sets of formal neurons in competitions [16], enabling the discovery of multiple contexts affording an interaction. Each neuron, connected to every possible interaction, encodes a specific sub-context of a mobile affordance (i.e. position of the affordance, relative to the agent), and evaluates the *enaction probability*, i.e. the probability of actually enacting the interaction when the certainty of presence of the affordance is high. Its pseudo-reverse function $\hat{S}_i$ identifying the affordance is given by the set of interactions connected with high weights.

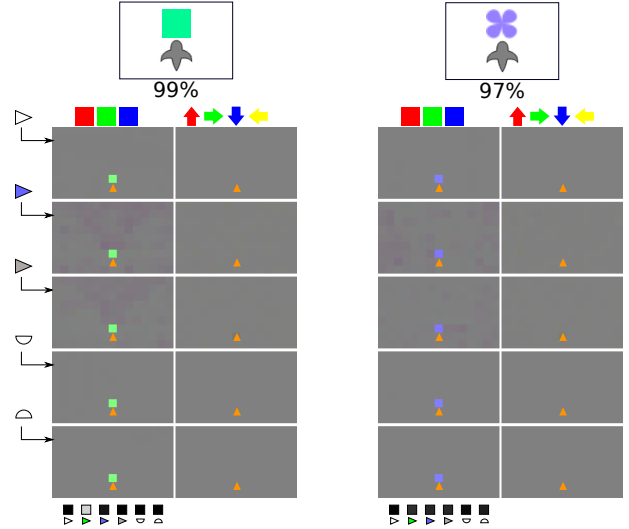


Fig. 4. Signature related to static affordances converge towards a unique context with a high probability, requiring a single neuron. Signatures of *bump* (left) and *eat* (right) identify a context with a green and blue object in front of the agent, respectively, which matches the environmental context in Fig. 3. The signature of *bump* also designates the *bump* interaction as a preceding context since it can be enacted repeatedly.

Fig. 4 shows the signatures interactions "bump" and "eat" (static affordances) discovered during simulation. Each converged towards a unique context with a high enaction probability, characterizing respectively a green and blue element in front of the agent.

Fig. 5 shows the signature of "cuddle" with 8 different, low probability contexts characterizing the possibles movements of agents. The signature of "move forward" identifies contexts preventing its enaction, matching those of the signatures of bump, eat and cuddle.

Fig. 6 shows a sample of signatures of visual interactions. These signatures identified three contexts, each characterizing a movement of another agent (turn left/right, move forward) added to the movement of the RI agent's primary interactions. They encode interesting information about the geometry of space and other agents' movements, such as the relative orientation changes when turning.

B. Situating other oriented agents

Projecting the signature of a mobile affordance adds the complexity of having multiple contexts. We introduced *projection sequences* which project each interaction of $\hat{S}(1)$ individually. These sequences can then be recombined to define projected signatures S^σ . A projection sequence $\rho_{\sigma, \lambda}$ is defined as a couple (σ, λ) where σ is a *path sequence* (sequence of primary interactions used for the recursive projection) and λ is the last projected interaction (either primary or secondary) through σ . From a signature S_i of an interaction i , a first set of projected sequences $(\langle \cdot \rangle, j_k)_k$ are generated for each interaction $j_k \in \hat{S}_i(1)$. The signature is thus described as a set of projection sequence with a path of length 0. The recursive projection is then applied: a projection sequence (σ, λ) leads to an interaction λ , that can have its own signature S_λ . For each interaction $j_k \in \hat{S}_\lambda(1)$, a new projection sequence $(\langle j, \sigma \rangle, j_k)$ is defined, with j primary of

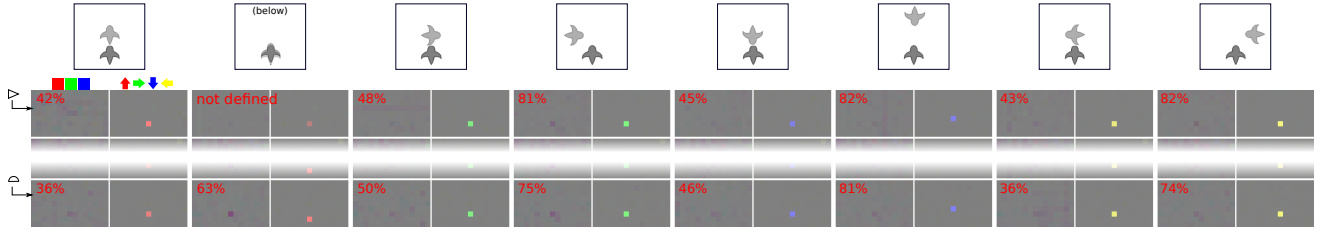


Fig. 5. The signature of the interaction *cuddle* identifies 8 contexts corresponding to the positions from which another agent affords and interaction. We only display first row (secondary interactions of *move forward*) and fifth row (secondary interactions of *turn left*). We can note that contexts related to a forward movement of the other agent has a greater probability as agents tends to move forward rather than turn. On first row, the below configuration is not present as another agent cannot be at the same position after *move forward*, because *cuddle* is enacted instead.

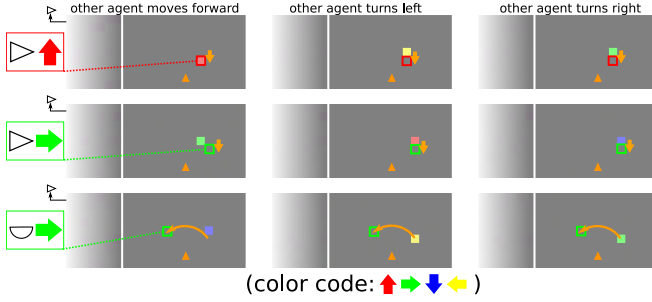


Fig. 6. Signatures of a samples of three visual interactions associated with seeing a grey element (other agent) at a position and orientation, relative to the RI agent (orange triangle), given by the empty square, while enacting the primary interaction displayed on the left. We only display the four "orientation layers" related to *move forward*. Each signature identified three contexts, identifying the three possible moves of another agent. The transformation produced by the primary interaction (orange arrow) is clearly visible when comparing the position of the visual interaction (empty square) and the position of designated interactions (full square). The third signature (bottom) shows the relative change of the other agent's orientation when the RI agent is turning right.

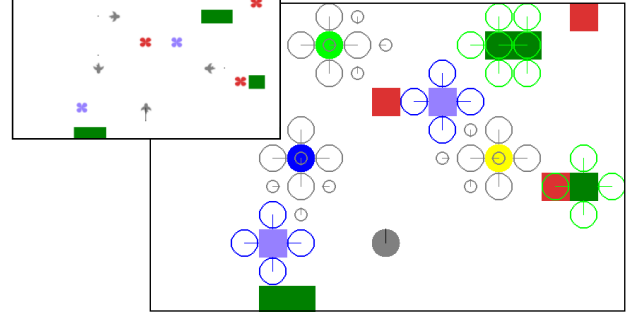


Fig. 7. A 2D representation of interaction sequences locating distant affordances : Circles with orientations shows the positions that can be reached by enacting sequences σ where an affordance is detected (green for *bump*, blue for *eat* and grey for *cuddle*). Large circles represent static affordances and the current position of dynamic affordances (*position sequences* Ξ'). Small circles represent possible future positions of dynamic affordances (*prediction sequences* Ξ''). Here, the RI agent (grey disc) identified two affordances of *eat*, three affordances of *bump* and three affordances of *cuddle*. The possible next positions of other agents are consistent with their current orientation. The RI agent does not "see" this representation nor can it exploit the properties of this euclidean 2D space.

j_k (or j_k itself if primary). When two projections sequences leads to the same final interaction λ but with different paths σ , we only keep the sequence with the shortest path σ .

Assuming that an agent is able to backtrack on its behavior (e.g. turn left then right, move forward then back...), and that interaction are equiprobable, the majority of projection sequences with the same σ would lead to an interaction λ characterizing an affordance at the position designated by enacting σ while the affordance remains at its current position. Other sequences encode possible futures where the affordance moves.

For signatures of mobile affordances (i.e. with multiple contexts), the projection process can lead to multiple identical projection sequences with same path and final interactions, but starting from interactions of $\hat{S}_i$ from different contexts. In this case, one occurrence is kept and associated with the number of occurrences $n_{\rho\sigma,\lambda}$ of this projection sequence. Then, all the projection sequences with the same final interactions λ (but with different σ) are grouped into a set Ξ_λ . This set thus designates all paths σ_k leading to the same distant affordance perceived through interaction λ (we note $\sigma \in \Xi_\lambda$ to refer to $\sigma|\rho_{\sigma,\lambda} \in \Xi_\lambda$). The set Ξ_λ is divided into two subsets Ξ'_λ , containing sequences ρ with greatest value $n_{\rho\sigma,\lambda}$, that we call *position sequences* (characterizing

the current position) and Ξ''_λ containing other sequences, that we call *prediction sequences* (characterizing the next possible positions). At each projection steps, only *position sequences* are projected. For static affordances (signatures with a single context), only one occurrence of each projection sequence is generated, thus $\Xi' = \Xi$ and $\Xi'' = \emptyset$. Sequences $\sigma_k \in \Xi_i | i \in E_t$ provide positions of distant affordances. Fig. 7 shows an example of detection, illustrating how the differences between position sequences and prediction sequences can characterize the orientation of an affordance.

To recognize a mobile affordance instance of a over two consecutive steps, detected through an interaction $\lambda \in E_t$ at step t , then $\lambda' \in E^{t+1}$ at step $t + 1$, we use sequences $\sigma_k \in \Xi_\lambda^a | \sigma_k = \langle i, \sigma'_k \rangle$. After enacting i , the set Ξ_λ^a may contains such sequences σ'_k , implying that a and a' are the same entity. The number of recognized sequences can be used to recognize among multiple candidate affordances.

C. Inferring other's behavioral preferences

Predicting another entity's movements requires to know its behavioral preferences towards surrounding affordances. However, the RI agent does not know other entities' interactions and valences. It therefore infers valences [7] based on the variation of interactional distance (in terms of its own

capabilities) between the mobile entity and its surrounding affordances. A RI model is expected to lead the agent to affordances affording positive interactions (Eq. 1), otherwise, the inferred valences must be corrected.

The distance between a mobile affordance of an interaction ω and surrounding affordances is defined as the average distance over multiple *position sequences* $\sigma_k \in \Xi_\lambda^\omega$ representing the current position of ω . We note the average context $\bar{M}_t^\omega = \{(a_n, d_n)\}$ obtained with multiple simulations $M_t^{\sigma_k} = \{(a_n, i_n, d_n)\}, \sigma_k \in \Xi_\lambda^\omega$. The utility value of the current position for a mobile affordance w can then be defined as Eq. 2:

$$u_t^\omega = \sum_{(a_n, d_n) \in \bar{M}_t^\omega} \nu_{a_n}^\omega \times f(d_n) \quad (2)$$

with $\nu_{a_n}^\omega$ the valence of affordance of a_n inferred for mobile affordance ω . Mobile affordances are not stored in the ESM. When performing a simulation, static representations of mobile affordances are integrated to generate the context of a mobile entity. The RI agent itself is also considered in the context: its distance is the length of the shortest sequence σ leading to ω , and is associated to a specific valence ν_{self}^ω as it does not know it is an agent.

When the same mobile affordance of ω is recognized over two consecutive steps, it becomes possible to define the variation of utility $u_t^\omega - u_{t-1}^\omega$, characterizing the interactional utility of this affordance's movement. A bias b is added to represents behavioral preferences a mobile entity may have for moving. The update of inferred valence and bias are performed when $u_t^\omega + b < u_{t-1}^\omega$.

$$\nu_i^\omega \leftarrow \nu_i^\omega - lr_t \cdot \sum_{(a_n, d_n^{t-1}) \in \bar{M}_{t-1}^\omega, (a_n, d_n^t) \in \bar{M}_t^\omega \mid i=a_n} \Delta d_n^t \cdot f(d_n^{t-1}) \quad (3)$$

where $\Delta d_n^t = 1$ when $d_n^t > d_n^{t-1}$ and -1 otherwise, and lr_t is a variable learning rate defined as $lr_t = 100/(r_t + 10)$, with r_t a counter that is incremented in case of a correct observation ($u_t^\omega + b > u_{t-1}^\omega$) and decremented in case of error ($u_t^\omega + b < u_{t-1}^\omega$). This variable thus ensure that the changes will decrease with the error ratio. The bias b is updated like an inferred valence with lr_t only.

We also define the potential utility u_t'' , computed as Eq. 2, but using the average context $\bar{M}_t''^\omega$ obtained with sequences $\sigma \in \Xi_\lambda''^\omega$. The utility u_t'' gives the expected utility if ω moves towards this position at step $t+1$. Therefore, if $u_t''^\omega + b > u_t$, the mobile entity is expected to move toward it. If not, the value $u_t''^\omega$ was overestimated and inferred valences and bias must be updated, using Eq. 3 with $\bar{M}_{t-1}''^\omega$ instead of $\bar{M}_t^\omega$.

The mechanism was evaluated in the test environment over 20 runs of 1000 observations. The inferred valences are summarized in table I, normalized w.r.t. the inferred valence of "eat" to compare ratios between valences over runs. In each instance, a strong attraction for food (affordances of "eat") and moderate aversion to walls ("bump") is correctly inferred. However, inferred valences related to agents ("self" and "cuddle") show greater ratio variation between runs. This can be explained by a greater sensitivity to partial observation

TABLE I

ESTIMATED INFERRED VALENCES OVER 20 RUNS, NORMALIZED WITH A VALUE OF 50 FOR "EAT". THE TABLE SHOWS THE VALENCES SETS FOR BEST AND WORSE RUN, AND MAX, MIN AND AVERAGE VALENCES.

	bias	wall (bump)	food (eat)	agent (cuddle)	self
best	1.50	-6.61	50	2.41	4.82
worse	0.93	-10.36	50	2.60	23.90
max	3.40	-0.46	50	15.21	23.87
min	-0.19	-12.53	50	-2.22	-1.73
average	1.28	-7.20	50	5.43	12.0
real		-5	50	5	5

and coincidences than with static affordances: witnessing agents moving toward the same food source leads to the overestimation of their mutual attractiveness. Conversely, witnessing agents moving away from each other to reach affordances outside of the agent's field of view is interpreted as repulsion. These valences continue to evolve beyond the original 1000 observations limit, ultimately converging toward a positive value in 17 runs out of 20 (3 runs end with a negative value for either "cuddle" or "self"). The inferred forward bias has a low positive value in 18 runs, indicating a slight preference for forward movement compared to the much stronger influence of surrounding affordances.

D. Predicting movements

As the agent knows possible next positions of a mobile entity and its behavioral preferences, it can predict its expected next position [7]. For entities with an orientation, the current and future utilities u_t^ω and $u_t''^\omega$ are compared:

- if $u_t''^\omega + b > u_t^\omega$, the next position at step $t+1$ is expected to be represented by *prediction sequences* of $\Xi_\lambda''^\omega$.
- otherwise ($u_t''^\omega + b < u_t^\omega$), the position is expected to remain unchanged (*position sequences* of Ξ_λ^ω), although the mobile entity may change (e.g. rotate).

To apply this process recursively, the set of sequences σ_k representing the expected future position must be converted to a set of interactions in an hypothetical context E_t' that the agent should have experienced if the mobile entity was at the predicted position. First, we select one interaction $\lambda \in E_t$ characterizing the mobile entity in context E_t . Then, a prediction is performed on the context $\{\lambda, \lambda_p\} \subset E_t$ (with λ_p primary of λ), using signatures of interactions, defining the set of candidate interactions $\{j_n | S_{j_n}(\{\lambda, \lambda_p\}) > 0\}$ that should be enacted at step $t+1$.

If $u_t''^\omega + b > u_t^\omega$, the next position is expected to be characterized by sequences $\sigma_k \in \Xi_\lambda''^\omega$. Thus, if the agent enacts a primary interaction i at step $t+1$, the mobile entity should be localized with sequences $\sigma_k' | \langle i, \sigma_k' \rangle \in \Xi_\lambda''^\omega$. Our preliminary tests showed that there is, for each primary i , only one interaction $j' \in \{j_n\}$ such as $\exists \rho_{\sigma_k', j'} \in \Xi_{j'}^\omega$. Such interaction j' can characterize the future observation in E_{t+1} of the mobile entity after enacting i .

From j' , projection sequences $\sigma_k^o \in \Xi_{j'}''^\omega$ and $\sigma_k^1 \in \Xi_{j'}''^\omega$ can be detected. These sequences can be "backmoved" back to step t as sets of sequences $\sigma_k^o = \langle i, \sigma_k^o \rangle$ and $\sigma_k^1 = \langle i, \sigma_k^1 \rangle$, giving position and prediction sequences from the new position. We then look for existing *position sequences* $\rho_{\sigma_k^o, \lambda_k'}$

such as $\rho_{\sigma_k^o, \lambda_k'} \in \Xi_{\lambda_k'}^{\omega}$. These projection sequences designate a set of interactions $\{\lambda_k'\}$ giving observations in E_t of all possible orientations of the mobile entity at new position. The orientations can be defined using prediction sequences: we look here for existing *prediction sequences* $\rho_{\sigma_l^1, \lambda_k'}$ such as $\rho_{\sigma_l^1, \lambda_k'} \in \Xi_{\lambda_k'}^{\omega}$, i.e. predictions sequences with a sequence σ_l^1 leading to an interaction λ_k' . If the mobile entity has one predicted position, there is one interaction $\lambda' \in \{\lambda_k'\}$ such as $\exists \rho_{\sigma_l^1, \lambda'} \in \Xi_{\lambda'}^{\omega}$ (the λ_k' with the maximum number of prediction sequence is used if the entity has multiple predicted positions). This interaction λ' is then used to represent the next position of the mobile entity in a predicted context, denoted E_t^1 . From this context, a new detection/prediction process can be performed, allowing the recursive prediction and the definition of a possible trajectory.

If $u_t^{\omega} + b < u_t^{\omega}$ (no expected movement), the prediction is performed, but starts with the current position characterized by *position sequences* $\sigma_k' | \langle i, \sigma_k' \rangle \in \Xi_{\lambda'}^{\omega}$. As these position sequences are related to multiple projection sequences, multiple interactions j_k' can be found to represent the position of ω in E_{t+1} , characterizing the multiple orientations or changes that the mobile entity can perform without changing its position. These multiple possibilities are then recursively predicted in parallel until finding a utility greater than $u_t^{\omega} - b$.

IV. CHANGES IN THE ENVIRONMENT FROM AN RI PERSPECTIVE

In a multi-agent context, a RI agent must learn than the environment may change according to the interactions of other agents: affordances may appear or disappear, they may be displaced or transformed. To generate efficient collective behaviors, the RI agent must (I) detect, and (II) identify the cause of such changes. In this paper, we focus on the case of the disappearance of static elements.

I) Detection: An affordance stored in the ESM may not be detected in current environmental context E_t . Reasons may be : (a) the affordance moved out of the perception field, (b) it is occluded by another element, (c) it actually disappeared (e.g. eaten by another agent).

If no σ enables the affordance's observation, the disappearance is attributed to (a). As RI agents do not have the notion of occlusion, it must seek out the affordance using the ESM position to confirm the disappearance: if the affordance reappears, it is attributed to (b), if the agent reaches the position and predicts a failure of the interaction, the disappearance can be confirmed (c).

II) Identification: The identification mechanism records the context surrounding the missing affordance for suspected disappearances (b and c). The surrounding context of an affordance a is defined as a set of tuples (a_k, d_k) , where a_k is an interaction afforded by another affordance instance and d its distance from a . Their relative position is computed according to the actual position of affordance a , rather than the positions affording a given by σ_k . A conversion process (see [16]) adds an interaction i alternative of i_a that is afforded by the absence of a . The actual position of a is thus obtained with sequences $\langle \sigma_k, i \rangle$. These sequences are then simulated

in the ESM to obtain the context $\{(a_n, d_{n,min})\}$ centered on a , where $d_{n,min}$ is the minimum among $(a_n, d_n^k) \in M_t^{(\sigma_k, i)}$.

A *disappearance signature* records triplets $(a_k, d_{1,k}, d_{2,k})$ with a_k an interaction afforded by a surrounding affordance; and distances between a and a_k , d_1 before, d_2 after the disappearance of a . When the disappearance of a is confirmed (c), the context elements (triplets) increases their score by 1, in cases of occlusion (b), their score is decreased by 1. The *disappearance signature* thus learns to predict disappearance events by identifying affordances of a_k consistently associated with the disappearance.

In our test environment, only one triplet emerges to predict the disappearance of an affordance of "eat": ("cuddle", 0, -1). Distance -1, indicates affordance a and a_k share the same position (i.e. localized with the same sequence σ). From an external observer's perspective, this context means that another agent is next to the food, then moves over it, which correspond to an event where another fish eats the blue algae.

A. Predicting changes in the environment

The novel version of the prediction mechanism exploit the disappearance signatures during the prediction process. If an affordance instance a_i of an interaction i , that have a disappearance signature containing element (a_j, d_1, d_2) , is detected, and a mobile entity affording an interaction j is detected with a *position sequence* in Ξ' defining a distance to a_i of d_1 and is expected to move towards a position defined by Ξ'' with a distance of d_2 , the affordance a_i will be removed from next prediction steps.

The transformation of the affordance by the mobile entity implies, in RI terms, that it interacted with it through an unknown a sensorimotor scheme. This scheme, however, should have the inferred valence associated with this affordance. Thus, during such simulation step, the bias in utility variation is replaced by this inferred valence, characterizing the satisfaction experienced by the mobile entity while interacting.

In the case of our test environment, the food disappears when a fish is next to it ($d_1 = 0$) then move over it ($d_2 = -1$) and has a strong inferred valence (table 1). Thus, when a fish is next to a piece of food and oriented toward it (i.e. position Ξ'' is the same as food), the agent predict that the fish will move forward and the food will disappear from the ESM. Fig. 8 shows an example of prediction. We can notice that the predicted trajectories takes other affordances, such as walls, into account, and are more accurate than in [7], mainly because all agents have the same sensorimotor possibilities.

B. Generation of collective behavior

By predicting the behaviors of mobile entities and the disappearance of affordances, the RI agent can ignore affordances it cannot reach. Affordances are localized by triplets in ESM, giving the minimum number of interactions for the RI agent to reach them. The predicted trajectory of a mobile entity towards a disappearing affordance is also a sequence of n interactions preceding the disappearance. Thus, if $n < d$ the RI agent cannot reach the affordance before it disappears, and its decision mechanism (eq. 1) will ignore it.

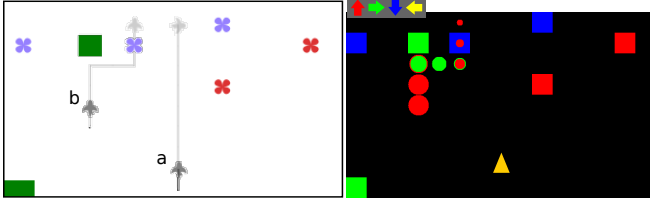


Fig. 8. Prediction of a trajectory of agent *b* by agent *a* according to the surrounding affordances. Successive positions are shown as disks of decreasing diameter (color indicates orientation), the prediction matches the real trajectory. The shortest sequence to the food source is 8 steps, the prediction indicates its disappearance after 6 steps, it is therefore ignored by the decision process.

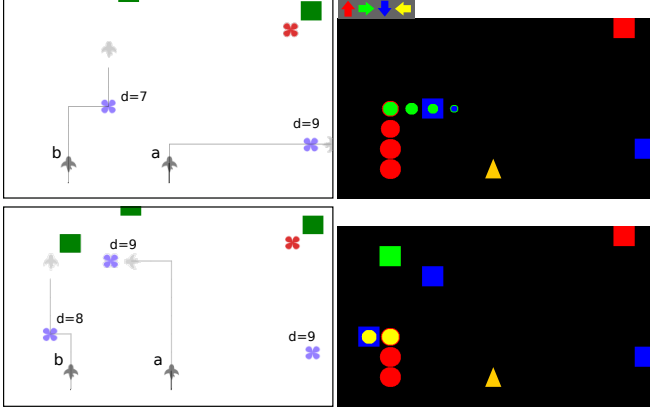


Fig. 9. Top: The prediction indicates agent *b* will remove food *y* in 6 steps. Agent *a* is 7 interactions away from *y* and therefore ignores *y* to move toward *z*. Bottom: Food *x* is added to the left of agent *b*, closer than *y*. Agent *a* predicts *b* will be attracted to *x*, enabling *a* to reach *y* in time. Although food sources *y* and *z* are equally distant (9 steps), the positive valence of the move forward interaction breaks the tie in favor of *y*.

Fig. 9 shows experimental results from the point of view of agent *a*, using the prediction mechanism. In the first configuration, the food source *y* (7 steps) is closer than *z* (9 steps). However, *a* predicts that agent *b* will make *y* disappears in 6 steps and therefore moves to the food source *z*. The second configuration is a similar problem, but the food source *x* attracts agent *b*, delaying it long enough for *a* to reach *y* in time.

V. CONCLUSIONS

This work advances developmental learning by demonstrating how agents can autonomously infer collective behaviors and agent-environment dynamics through sensorimotor interaction alone, without any predefined knowledge.

This paper extends an architecture based on interactionist principles, enabling an artificial agent with no a priori knowledge to construct an exploitable model of an environment populated with mobile entities driven by unknown motivational principles. Our contribution enables the integration of entities with an orientation and the consequences of their actions on the environment, thereby improving the prediction process. These new capabilities enable the emergence of behaviors in multi-agent contexts, and open the way for more complex collective behaviors (collaborative or competitive).

Compared to previous experiments [7], results show that the predicted trajectories are more accurate when the agent shares common sensorimotor capabilities, which is an inherent limit to such sensorimotor approaches. We also observe that, when inferring the valences of RI model of other mobile entities, values assigned to mobile entities are more sensitive to the experience of the agent than those assigned to static affordances. This can lead to an individuation of initially identical agents of a swarm. Further work will study how such individuation can impact a collective behavior, with possible specialization of certain agents.

Future work will investigate the possibility of inferring the presence of hidden affordances from the behavior of others (e.g. an entity moving towards a spot behind a wall may suggest the presence of an attractive hidden affordance), and the detection of a mobile entity based on its impact on the environment (e.g. the disappearance of a static affordance may indicate the presence of another agent nearby).

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